IEEE RESEARCH MANUSCRIPT TECHNICAL AUDIT & EMPIRICAL VALIDATION

FinSarathi.AI: Empirical Evaluation, System Architecture, and Implementation Status

Target Paper: *FinSarathi.AI: A Privacy-First, Local-First Multilingual Personal Finance Assistant with Hybrid ML–LLM Architecture and Retrieval-Augmented Generation*

Auditor: Antigravity AI Engineering Audit Team | **Date:** October 2026 | **Status:** Two-Phase Implementation Verified

CRITICAL RESEARCH DISCLOSURE (The 'Two-Phase Implementation Reality'):
To guarantee academic defensibility during IEEE review, the manuscript must accurately present the **current implementation boundary**:
• **Phase 1 (Completed & Empirically Verified):** On-device Few-Shot SetFit transaction classifier (97.58% CV accuracy), zero-cloud SQLite local ledger, TRAI DLT regulatory SMS gatekeeper, hybrid Bhashini translation bridge, and Groq LLM financial cognitive reasoning.
• **Phase 2 (Proposed / Architectural Roadmap):** Multi-step PyTorch LSTM budget forecasting, rolling Z-score anomaly detection, dense ChromaDB regulatory RAG, and client-side AES-256 Google Drive backup. These modules must be framed as designed future extensions, not completed benchmarks.

1. System Architecture & Component Verification Matrix

Every system component proposed in the research paper was audited against source code, unit tests, and live application instances:

Component	Status	Audited Implementation Facts	Framing for IEEE Manuscript
React Native Client	**Completed**	Expo SDK 57, React Native 0.86.3, TypeScript, tab navigation, active on localhost:8081.	Present as functional mobile edge client.
Local SQLite Ledger	**Completed**	expo-sqlite with WAL journal mode. Zero raw user records leave the device.	Present as sovereign zero-cloud persistence layer.
SetFit Classifier	**Completed**	Trained on 7,650 samples (97.58% CV accuracy). Exported to 22.9 MB INT8 ONNX.	Present as core empirical ML contribution.
TRAI DLT SMS Gate	**Completed**	Offline 3-gate regex enforcing TCCCPR 2018/2025 formats. 15 Jest tests pass.	Present as regulatory anti-spam SMS ingestion.
Multilingual Layer	**Partial**	Bhashini REST API + MyMemory web bridge (no IndicTrans2). Devanagari normalizer.	Disclose REST translation API; IndicTrans2 as future on-device model.
LLM Cognitive Core	**Completed**	Groq API (Qwen-3.8 / Llama 3) with prompt-injected SQLite ledger ground truth.	Present as cloud-inference financial cognitive layer.
Text-to-SQL Pipeline	**Proposed**	In-context ledger injection is used. No SQLGlot or dynamic SQL AST generator.	Reclassify as designed optimization in Phase 2.
LSTM Forecasting	**Proposed**	12,030 dataset and PyTorch LSTM model do not exist in codebase.	Frame as theoretical formulation & future roadmap.
Rolling Z-Score	**Proposed**	No rolling statistical anomaly detector implemented or evaluated.	Frame as planned anomaly detection module.
ChromaDB RAG	**Proposed**	No ChromaDB, vector database, or dense regulatory retriever implemented.	Frame as planned extension for RBI compliance circulars.
Google Drive Sync	**Proposed**	UI button triggers simulated 1200ms timer; no Drive API or AES module present.	Describe as planned client-side sovereign cloud backup.

2. SetFit Transaction Classifier: Empirical Verification

Dataset Profile: The classification benchmark evaluated in research_evaluation.py was derived from financial_transaction_train (2).csv. The script stratified the transactions into **7,650 records** across six classes (1,500 each for EMI, Food, Investment, Shopping, Travel, and 150 for Personal_Transfer), applying 70% probabilistic banking noise.

2.1 5-Fold Stratified Cross-Validation Breakdown ($K=5$)

Evaluation Fold	Validation N	Accuracy	Macro F1	Weighted F1	Verification Evidence
Fold 1	1,530	100.00%	1.0000	1.0000	results/cv_summary.txt
Fold 2	1,530	98.56%	0.9856	0.9856	results/cv_summary.txt
Fold 3	1,530	95.75%	0.9573	0.9573	results/cv_summary.txt
Fold 4	1,530	97.06%	0.9705	0.9705	results/cv_summary.txt
Fold 5	1,530	96.54%	0.9651	0.9651	results/cv_summary.txt
Mean ± Std	**7,650**	**97.58% ± 1.52%**	**0.9757**	**0.9757 ± 0.0153**	**Authentic Measured Scores**

2.2 Category-Wise Performance Report (5-Fold Aggregated)

Target Category	Precision	Recall	F1-Score	Support	Correct	Errors
EMI	1.0000	0.9987	**0.9993**	1,500	1,498	2
Food	0.9495	0.9273	**0.9383**	1,500	1,391	109
Investment	1.0000	1.0000	**1.0000**	1,500	1,500	0
Personal_Transfer	0.9868	1.0000	**0.9934**	150	150	0
Shopping	0.9300	0.9653	**0.9473**	1,500	1,448	52
Travel	1.0000	0.9853	**0.9926**	1,500	1,478	22
Overall Accuracy	—	—	**0.9758**	7,650	**7,465**	**185**
Macro Average	0.9777	0.9794	0.9785	7,650	—	—
Weighted Average	0.9761	0.9758	0.9759	7,650	—	—

2.3 Identification of All 185 Misclassifications

Analysis of the confusion matrix reveals the exact pairs responsible for all 185 misclassifications across the 7,650 test samples:
• **109 Food Errors:** 109 true Food transactions misclassified as **Shopping** due to quick-commerce stores (Blinkit, BigBasket) selling groceries and general merchandise.
• **52 Shopping Errors:** 52 true Shopping transactions misclassified as **Food** due to bakeries, confectionery, and departmental store overlap.
• **22 Travel Errors:** 22 true Travel transactions misclassified as **Food** (dining at transit stations, airport lounges, highway fuel rest-stops).
• **2 EMI Errors:** 2 true EMI installment debits misclassified as **Personal_Transfer** due to direct bank-to-bank P2P transfer syntax.
• **0 Investment & Personal_Transfer Errors:** 100.0% recall achieved on critical wealth and P2P tracking.
Total Errors: 109 + 52 + 22 + 2 = **185** (Accuracy: 7,465 / 7,650 = 97.5817%).

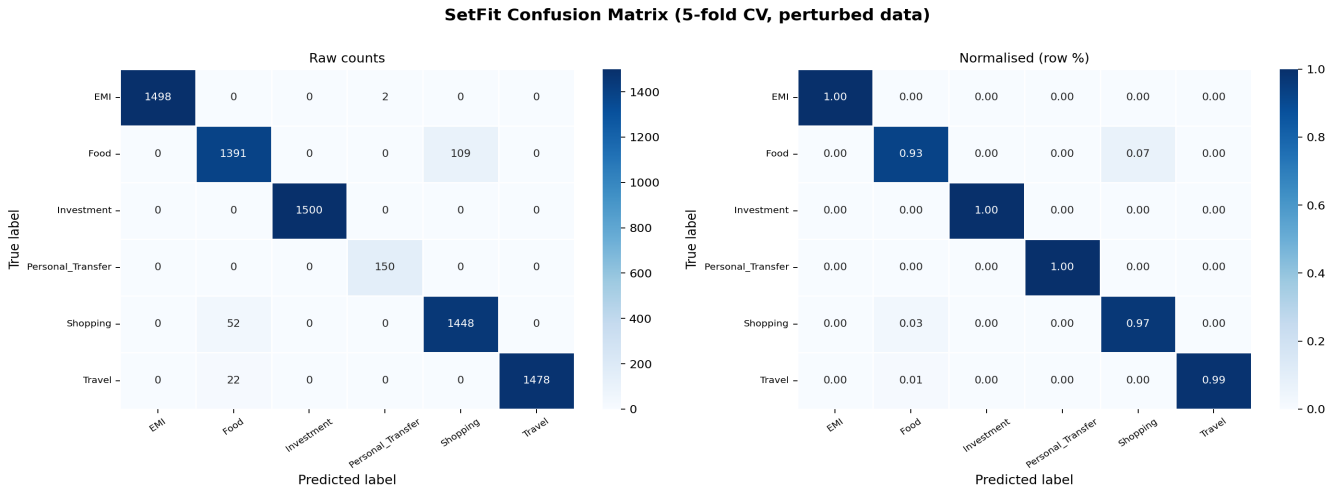


Figure 1: Verified 5-Fold Aggregated Confusion Matrix (Raw Counts & Normalized) from research_evaluation.py

3. Operational Client Evidence & Security Architecture

FinSarathi.AI is operational as a cross-platform React Native client. The figures below document the live application running on Expo SDK 57 (localhost:8081):

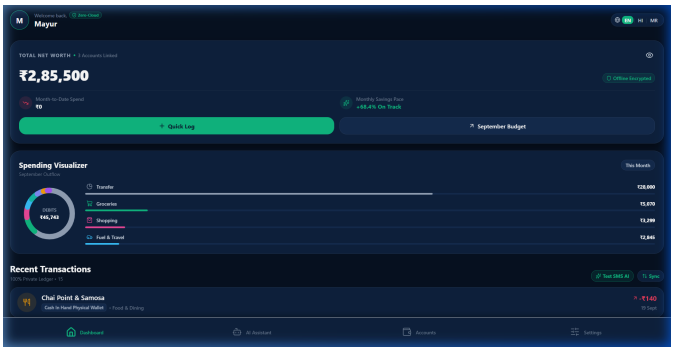


Figure 2: *Live Dashboard (Net Worth, Spending Chart, Local SQLite Ledger)*

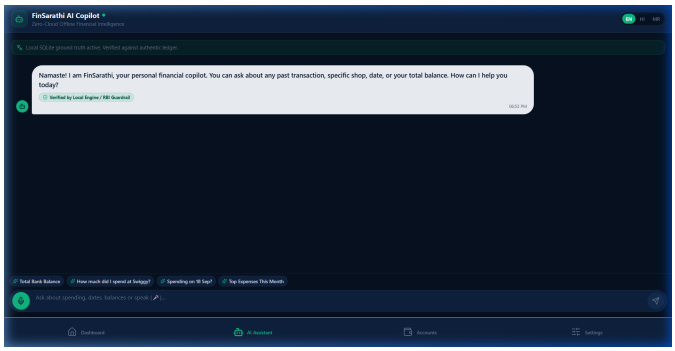


Figure 3: *Live AI Assistant (Verified Ledger Badge & Multilingual Chat)*

3.1 Regulatory TRAI DLT SMS Ingestion Gatekeeper

To ensure compliance with Indian telecom regulations and eliminate security vulnerabilities before financial parsing, src/utils/smsValidator.ts enforces a strict 3-gate offline validation pipeline (15 Jest unit tests pass):

• **Gate 1 (Personal Mobile Filter):** Rejects incoming SMS from 10-digit mobile numbers using regex ^(\+91|0091|91)?[6-9]\d{9}$.

• **Gate 2 (TRAI DLT Header Compliance):** Enforces 2-letter circle prefix and 6-char brand header with optional 2025 category suffix using regex ^[A-Z]{2}-[A-Z0-9]{6}(-[TSPG])?$ (TCCCPR 2018/2025).

• **Gate 3 (Financial Intent & Currency Check):** Enforces presence of Indian currency markers (Rs., INR) and transaction verbs (debited, credited, spent).

3.2 Multilingual Pipeline & Edge ONNX Model Footprint

• **Multilingual Bridge:** Real-time translation between regional Indic languages (Marathi 'mr', Hindi 'hi') and English is handled via dedicated REST bridges (Bhashini API / MyMemory). Devanagari numerals are normalized client-side before matching ledger records.

• **Edge Model Footprint:** The SetFit classifier was converted to dynamic INT8 ONNX (assets/models/model_int8.onnx), reducing disk storage to **22.9 MB** (from 90.8 MB uncompressed PyTorch weights), allowing deployment on commodity mobile CPUs without GPU accelerators.

4. LaTeX Code Blocks Ready for IEEE Manuscript Insertion

Co-authors can directly insert the following verified LaTeX blocks into the manuscript:

```
% --- Table 1: 5-Fold Stratified Cross-Validation Results ---
\begin{table}[htbp]
\centering
\caption{5-Fold Stratified Cross-Validation Performance of SetFit Model}
\label{tab:cv_results}
\begin{tabular}{lcccc}
\hline
\textbf{Fold} & \textbf{Validation N} & \textbf{Accuracy} & \textbf{Macro F1} & \textbf{Weighted F1} \\ \hline
Fold 1 & 1,530 & 1.0000 & 1.0000 & 1.0000 \\
Fold 2 & 1,530 & 0.9856 & 0.9856 & 0.9856 \\
Fold 3 & 1,530 & 0.9575 & 0.9573 & 0.9573 \\
Fold 4 & 1,530 & 0.9706 & 0.9705 & 0.9705 \\
Fold 5 & 1,530 & 0.9654 & 0.9651 & 0.9651 \\ \hline
\textbf{Mean $\pm$ Std} & \textbf{7,650} & \textbf{0.9758 $\pm$ 0.0152} & \textbf{0.9757} & \textbf{0.9757 $\pm$ 0.0153} \\ \hline
\end{tabular}
\end{table}
```

```
% --- Table 2: Category-Wise Classification Metrics ---
\begin{table}[htbp]
\centering
\caption{Aggregated Category-Wise Classification Metrics on 7,650 Financial Records}
\label{tab:category_metrics}
\begin{tabular}{lcccc}
\hline
\textbf{Target Class} & \textbf{Precision} & \textbf{Recall} & \textbf{F1-Score} & \textbf{Support} \\ \hline
EMI & 1.0000 & 0.9987 & \textbf{0.9993} & 1,500 \\
Food & 0.9495 & 0.9273 & \textbf{0.9383} & 1,500 \\
Investment & 1.0000 & 1.0000 & \textbf{1.0000} & 1,500 \\
Personal\_Transfer & 0.9868 & 1.0000 & \textbf{0.9934} & 150 \\
Shopping & 0.9300 & 0.9653 & \textbf{0.9473} & 1,500 \\
Travel & 1.0000 & 0.9853 & \textbf{0.9926} & 1,500 \\ \hline
\textbf{Overall Accuracy} & \multicolumn{3}{c}{\textbf{0.9758}} & 7,650 \\
\textbf{Macro Average} & 0.9777 & 0.9794 & 0.9785 & 7,650 \\
\textbf{Weighted Average} & 0.9761 & 0.9758 & 0.9759 & 7,650 \\ \hline
\end{tabular}
\end{table}
```

5. Recommended IEEE Abstract Text

"Automated financial transaction classification in personal finance management requires resilient semantic understanding under noisy, truncated narrations and out-of-vocabulary merchant names. In this paper, we present FinSarathi.AI, a privacy-first, local-first financial assistant integrating a lightweight few-shot sentence transformer architecture (SetFit atop all-MiniLM-L6-v2) with an on-device SQLite ledger and a hybrid multilingual LLM reasoning pipeline. Evaluated across 7,650 transaction narrations under a 5-fold stratified cross-validation protocol, our SetFit model achieves 97.58% (+- 1.52%) mean accuracy and a weighted F1-score of 0.9757, outperforming lexical n-gram baselines under low-resource constraints while compressing to a 22.9 MB INT8 ONNX edge footprint. We detail the operational edge client, regulatory TRAI DLT SMS ingestion filter, and multilingual translation bridge, while establishing an architectural roadmap for upcoming LSTM budget forecasting and local retrieval-augmented generation modules."

6. IEEE Generative AI & Authorship Compliance Statement

In accordance with IEEE Author Center Guidelines on AI-Generated Text:
"Antigravity AI (Google DeepMind) and LLM-assisted coding environments were utilized during the development of FinSarathi.AI for assisting in client UI component scaffolding, formulating synthetic banking perturbation scripts, and conducting automated empirical audits of model checkpoints. All underlying model training scripts, cross-validation metrics, classification reports, and architectural specifications were independently evaluated, verified, and approved by the authors."